

# False Positive Validation & Detection Tuning

Sanitized one-page case study | Defender for Office 365 + KQL | Spring 2026

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## Context

Several emails quarantined as phishing were ultimately determined to be legitimate communications. Cases included Microsoft Store notifications, Canvas/Instructure emails, political fundraising campaigns, and educational digest platforms.

## Goal

- Validate whether detections were truly malicious or false positives.
- Reduce unnecessary disruption caused by aggressive automated detections.
- Use behavioral and infrastructure evidence instead of reputation verdicts alone.

## Key Actions

- Reviewed SPF/DKIM/DMARC authentication alignment.
- Validated infrastructure ownership and sender reputation.
- Compared sandbox behavior against URL reputation-engine verdicts.
- Correlated delivery telemetry and user context before release decisions.
- Submitted validated clean samples back to Microsoft for detection improvement.

## Results (sanitized)

- Released multiple legitimate emails incorrectly quarantined as phishing.
- Confirmed legitimate Microsoft, Canvas, and ActionNetwork infrastructure behavior.
- Reduced unnecessary remediation and false positive escalation.
- Improved operational understanding of detection-engine limitations.

## What this shows

I can perform evidence-based false positive validation, balancing security enforcement with operational continuity while supporting detection tuning and remediation workflows.